Section 4.1 — Exponential Functions — Answer Key

Math 103: College Algebra

Identifying Exponential Expressions

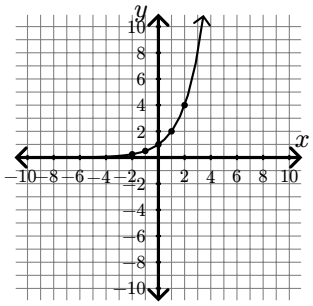
Determine whether or not each expression is exponential.

1. Yes
2. No
3. Yes
4. Yes
5. No

Graphing Exponential Functions

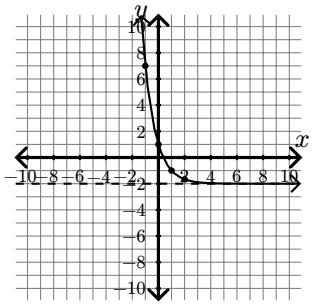
For each exponential function, find the points on the graph for each input (not x-value) in $\{-2, -1, 0, 1, 2\}$, and find the equation of the horizontal aysmptote. Then, sketch a graph of the function in the space provided. (If the point can't be plotted in this space, leave it off. Do your best to approximate fractions.)

6.



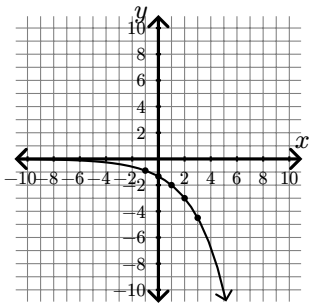
x	input	y
-2	-2	$\frac{1}{4}$
-1	-1	$\frac{1}{2}$
0	0	1
1	1	2
2	2	4

7.



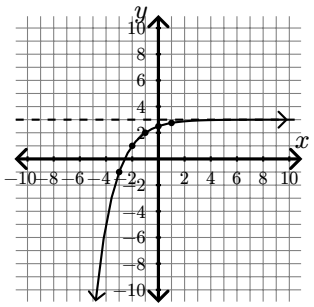
x	input	y
-2	-2	25
-1	-1	7
0	0	1
1	1	-1
2	2	$-\frac{5}{3}$

8.



x	input	y
-1	-2	$-\frac{8}{27}$
0	-1	$-\frac{4}{9}$
1	0	-2
2	1	-3
3	2	$-\frac{9}{2}$

9.



x	input	y
-3	-2	-1
-2	-1	1
-1	0	3
0	1	$\frac{5}{2}$
1	2	$\frac{11}{4}$

Identifying Graphs of Exponential Functions

Match each function to its graph, and give the equation of its horizontal asymptote.

10. c; $y = -\frac{1}{3}$
11. a; $y = 1$
12. d; $y = -\frac{3}{2}$
13. b; $y = -1$

Domain and Range of an Exponential Function

Find the domain and range of each exponential function.

14. $\operatorname{Dom} f = (-\infty, \infty)$
    $\operatorname{Rng} f = (-1, \infty)$
15. $\operatorname{Dom} g = (-\infty, \infty)$
    $\operatorname{Rng} g = (-\infty, -11)$
16. $\operatorname{Dom} h = (-\infty, \infty)$
    $\operatorname{Rng} h = (-\infty, 1)$
17. $\operatorname{Dom} k = (-\infty, \infty)$
    $\operatorname{Rng} k = (623, \infty)$

Exponential Equations

Solve each equation.

18. $x = 4$
19. $x = \frac{5}{3}$
20. $x = -\frac{3}{2}$
21. $x = -1, 3$
22. $x = -12$
23. $x = -4$
24. $x = 1, 3$
25. $x = 2, 4$